



UvA-DARE (Digital Academic Repository)

Exquisitor at the Video Browser Showdown 2025

Unifying Conversational Search and User Relevance Feedback

Sharma, Ujjwal; Khan, Omar Shahbaz; Rudinac, Stevan; Jónsson, Björn Þór

DOI

[10.1007/978-981-96-2074-6_31](https://doi.org/10.1007/978-981-96-2074-6_31)

Publication date

2025

Document Version

Final published version

Published in

MultiMedia Modeling

License

Article 25fa Dutch Copyright Act (<https://www.openaccess.nl/en/policies/open-access-in-dutch-copyright-law-taverne-amendment>)

[Link to publication](#)

Citation for published version (APA):

Sharma, U., Khan, O. S., Rudinac, S., & Jónsson, B. Þ. (2025). Exquisitor at the Video Browser Showdown 2025: Unifying Conversational Search and User Relevance Feedback. In I. Ide, I. Kompatsiaris, C. Xu, K. Yanai, W.-T. Chu, N. Nitta, M. Riegler, & T. Yamasaki (Eds.), *MultiMedia Modeling: 31st International Conference on Multimedia Modeling, MMM 2025, Nara, Japan, January 8–10, 2025 : proceedings* (Vol. V, pp. 264-271). (Lecture Notes in Computer Science; Vol. 15524). Springer. https://doi.org/10.1007/978-981-96-2074-6_31

General rights

It is not permitted to download or to forward/distribute the text or part of it without the consent of the author(s) and/or copyright holder(s), other than for strictly personal, individual use, unless the work is under an open content license (like Creative Commons).

Disclaimer/Complaints regulations

If you believe that digital publication of certain material infringes any of your rights or (privacy) interests, please let the Library know, stating your reasons. In case of a legitimate complaint, the Library will make the material inaccessible and/or remove it from the website. Please Ask the Library: <https://uba.uva.nl/en/contact>, or a letter to: Library of the University of Amsterdam, Secretariat, P.O. Box 19185, 1000 GD Amsterdam, The Netherlands. You will be contacted as soon as possible.



Exquisitor at the Video Browser Showdown 2025: Unifying Conversational Search and User Relevance Feedback

Ujjwal Sharma¹(✉)() , Omar Shahbaz Khan²() , Stevan Rudinac¹() ,
and Björn Þór Jónsson²()

¹ University of Amsterdam, Amsterdam, Netherlands
{u.sharma,s.rudinac}@uva.nl

² Reykjavik University, Reykjavik, Iceland
{omark,bjorn}@ru.is

Abstract. Exquisitor is a multimedia retrieval system designed to combine conversational search and user relevance feedback to enhance interactive video search. In this work, we present recent improvements to Exquisitor, focusing on integrating these two search strategies into a unified, user-friendly interface. In this system edition set to participate at VBS 2025, we propose improvements for fusing results from the conversational search and user relevance feedback subsystems into a single search pane, an ability to launch multiple tabbed search instances and several improvements to the user interface to significantly improve user experience and search efficiency, particularly for novice users.

Keywords: Interactive learning · Video browsing · Multimedia retrieval · Multimodal representation learning · Multimodal conversational search

1 Introduction

Interactive retrieval is a field of study that seeks to develop search systems that allow users to navigate and retrieve relevant information from large, complex datasets through active user engagement. With the rapid growth of multimedia content, traditional search methods often fall short in handling complex, ambiguously-worded or incomplete queries. Interactive search addresses this by allowing users to iteratively refine results through feedback, improving relevance of retrieved results. This approach is particularly valuable for tasks requiring detailed exploration, such as large-scale multimedia retrieval, where user input and system responses work together to bridge the gap between computational models and human intent.

Seeking to further the advancement of such systems, Video Browser Showdown (VBS) is an annual live benchmarking competition for interactive video search systems, aimed at evaluating their ability to retrieve relevant videos and associated information from a vast collection containing diverse semantic content

[10, 11, 16]. Held annually since 2012 at the International Conference on Multimedia Modeling (MMM), VBS traditionally comprises three distinct search tasks: Visual Known-Item Search (KIS-V), Textual Known-Item Search (KIS-T), Ad-Hoc Video Search (AVS), and Question and Answering (Q&A).

In KIS-V tasks, participants must locate a specific video using a textual description that is progressively expanded during the round to provide increasingly precise information. KIS-T tasks require retrieving the video that contains the necessary information to answer a given text-based query. In AVS tasks, systems are tasked with retrieving all videos that satisfy a particular search query, which tests the system’s recall capabilities. Q&A tasks present a video segment with a question relating to various parts of the full video. Thus, the objective is to locate the video and browse through it to find the answer. These results are submitted to the DRES evaluation server [13], where they are automatically or manually judged for relevance, depending on the task.

To evaluate the transferability of these systems across a range of visual search tasks, VBS uses a large dataset V3C, which contains 28450 diverse videos with a total duration of 3801 h [14]. Additionally, the Marine Video Kit (MVK 2.0) dataset [17], comprising visually similar videos captured by moving cameras in underwater environments (e.g., scuba diving), and the LapGynLHE dataset with surgeries in laparoscopic gynaecology, are used to test performance in more specific domains.

To ensure that the systems are usable by a wide range of users, VBS is divided into “expert” and “novice” stages. In the expert stage, the system is operated by its designers. In the novice stage, two unassisted novice users are asked to use the system after a brief tutorial, allowing the designers to assess the system’s usability and adaptability.

Our video search system, Exquisitor [5], has participated in five editions of the Video Browser Showdown (VBS) from 2019 to 2024, with the exception of 2023. Over these years, Exquisitor has evolved from an interface that only supported User Relevance Feedback (URF) to a hybrid system that incorporates a combination of conversational text search, URF, and traditional filters for search tasks. Additionally, significant efforts have been made to modernize the original monolithic codebase, transitioning to a simpler, modular system that is more accessible for new contributors. In the following sections, we introduce Exquisitor and outline the proposed changes for its latest version.

2 Exquisitor: System Overview

Exquisitor is a multimedia retrieval and analytics system designed to support large-scale interactive search by combining conversational text search with user relevance feedback (URF). In conversational search, users input queries to specify their needs and engage in multiple interaction rounds to iteratively refine the search results, while URF (User Relevance Feedback) involves users providing feedback on the system’s retrieval results, enabling the system to adjust its model and improve future search outcomes. This feedback is given by labeling

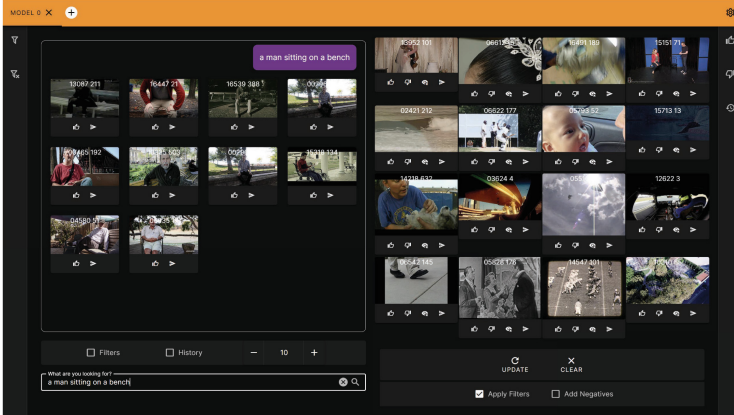


Fig. 1. Exquisitor’s split interface with Conversational Search (left) and User Relevance Feedback (right).

any number of media items as relevant or irrelevant [15]. Additionally, filters are used to narrow down the search space during tasks (Fig. 1).

The search system as a whole has undergone several revisions. Prior to 2021, Exquisitor was built exclusively around URF [3, 4]. While URF is itself an excellent search and exploration strategy, seeding the URF system with relevant positive examples was found to be difficult. Initially, a tag-based search system was developed to simplify this process by using a large number of concept annotations such as semantic concepts and actions. However, this was found to be largely insufficient as certain search queries contained complex semantics that were difficult to express via a limited set of concepts. To lift this restriction, Exquisitor at the Video Browser Showdown 2022 [6] introduced a rudimentary version of free text search built upon CLIP embeddings [1, 2, 12], which have proven to be a go-to method for many systems [10]. While this was found to be useful for the intended purpose, the overall process of embedding the queries in real time and matching it against the collection was exceedingly slow. To alleviate these concerns and to modernize the aging codebase, we rewrote most of Exquisitor’s backend from scratch for Video Browser Showdown 2024 [8]. The new system implements different search tasks as pluggable extensions into a modular backend server. This has allowed us to modernize and simplify the backend server, increase its reliability and open it up to improvements from contributors who might’ve found the original codebase troublesome to navigate.

3 Proposed Improvements

With the backend improvements complete, our focus shifts towards enhancing the user interface to facilitate more effective communication between users and the system. The demonstration of Exquisitor at the Interactive Video Retrieval

for Beginners (IVR4B) special session, part of the International Conference on Content-based Multimedia Indexing 2024, revealed an interesting insight: novice users still tend to favor textual search, despite half of the interface being allocated to URF. This strong preference for textual search is understandable, as users are generally more accustomed to it as the primary method in most search systems. Observations from IVR4B and Lifelog Search Challenge 2024 [7] indicate that users only resort to URF after spending significant time with textual search and failing to locate the relevant media items. Given the time constraints, users often revert to the approach they are most familiar with.

We conjecture that if users had more time to familiarize themselves with the system and explore alternative search methods, they would develop a better understanding of which tasks are better suited for URF compared to textual search or at least become less hesitant to use URF. While it would be ideal for users to have more time to interact with the system, the separation between the two approaches has maintained a certain degree of division between them. In Exquisitor’s latest version, conversational search and URF were combined, but the choice of method was still left to the user.

In this paper, we propose a more integrated approach, blending conversational search and URF into a unified workflow. The process begins with the user submitting a natural language query, followed by providing feedback on the results, which refines the search and retrieves a new set of results. From this point, the user can either continue giving feedback or adjust their query. This seamless integration encourages users to work with both methods naturally, without having to choose between them. Beyond this major change in Exquisitor’s approach, we also explore new interface designs aimed at improving video summary interactions and the ability to use multiple models within a session. We detail these improvements in the following sections.

3.1 Fusing User Relevance Feedback with Conversational Search

Exquisitor employs a user-mediated hybrid search strategy that combines results from both conversational search and user relevance feedback (URF). Conversational search can either be used independently to perform searches or to generate positive examples that seed the URF process. Negative examples are generally easier to identify, as most of the collection, apart from the positives, can be treated as potential negatives. In their role as intermediaries, users select positive and negative examples to guide the URF system. Based on the retrieved search results, users then decide how to adjust the query to refine and improve subsequent searches. In this strategy, while the results from conversational search influence the URF search through the selection of positive examples, the relationship is not reciprocal – URF results do not influence or transfer back to the conversational search process. This one-way interaction often results in a fragmented user experience, where Exquisitor appears as two separate systems rather than a unified hybrid system with synergistic components.

To address these issues, we propose an integrated hybrid search strategy in which both components – conversational search and URF – can influence

each other. In the current model, conversational search contributes to URF by providing potential positive examples. In the proposed system, we will introduce a reciprocal process where conversational search results are conditioned by the outcomes of the URF process. This integration will be achieved using rank aggregation between the top results of the two approaches. In this way, both the URF model and conversational search model can influence each other, creating a more cohesive search experience. A single-turn variation of this approach has been shown before at Video Browser Showdown with the SOMHunter system, where URF was used to refine the top results of the textual search [9].

3.2 Multiple Models Revisited

A core feature of Exquisitor – the ability to instantiate multiple models (search sessions in varying states of completion) – was introduced at Exquisitor’s run for VBS 2021. At a conceptual level, this is similar to having multiple search tabs, each holding a different search state. Users can initiate distinct search sessions, each with a unique search state, which can later be revisited or merged for improved results. During the backend rewrite, this feature was set aside as it required enablement on the new backend. However, we are now bringing this feature back to allow users the ability to open multiple search sessions with minimal resource footprint on the frontend. Additionally, interactions for all models will be logged by the logging subsystem introduced in Exquisitor’s run at IVR4B at CBMI 2024.

4 User Interface

Model Page (Main): Our proposed improvement aims to fuse the results from both conversational search and user relevance feedback (URF), eliminating the need to divide the interface into separate sections. This allows for a more streamlined, chat-style interface, as shown in Fig. 2. In addition to providing the best results based on rank aggregation from both approaches, we will also offer options to display top items from each method individually, along with a selection of diverse items from the collection. These features give users valuable insights into the performance of each approach for the active task. For example, in Ad-hoc Video Search tasks, relevant items can often be found more quickly using the URF approach. The diverse result set offers users opportunities for exploration or helps them identify better items to label as negatives. Ultimately, these result options serve as advanced features that expert users can leverage, yet they remain simple enough for novice users to benefit from as well.

Video Summary: At CBMI 2024, the IVR4B special session was held, during which two novice users employed Exquisitor to solve Visual Known-Item Search (KIS-V) and Q&A tasks on the VBS dataset (V3C1+V3C2). Throughout the session, we logged several instances where users interacted with the correct video but struggled to locate the precise segment needed to complete their task. These

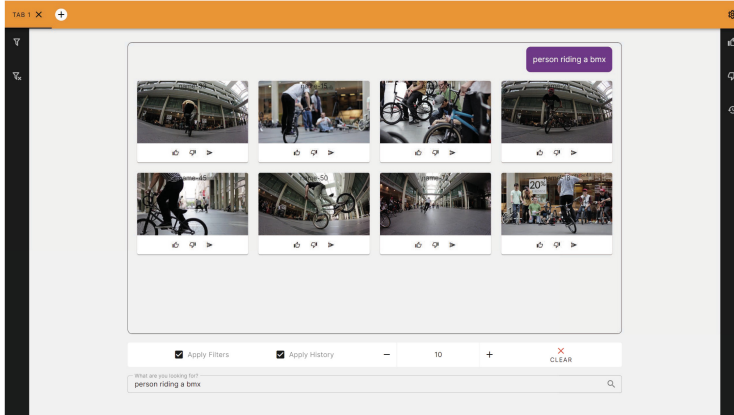


Fig. 2. The simplified UI for conversational and user relevance feedback search. Conversational search is positioned as the primary interaction method with the ability to invoke user-relevance feedback as a search refinement strategy.

difficulties were linked to limitations in Exquisitor’s video summary browser, which allows users to browse and play video segments. Since users could not view all the segments at once, they lacked a clear sense of the video’s total length and their current position within it. Furthermore, the playing segment was not highlighted, which contributed to additional confusion and time loss.

To address these issues, we have completely redesigned Exquisitor’s video summary browser. The updated browser now plays the entire video with options to loop selected segment(s) and adjust playback speed. This revision also resolves a recurring problem with submitting results to the DRES evaluation server. Previously, there were challenges in accurately sending the start and end timestamps for video segments, as they sometimes missed the target by half a second. The new system now enables the submission of precise timestamps chosen by the user, ensuring accurate reporting and improved task completion.

5 Conclusion

In this paper, we present a new version of Exquisitor that integrates conversational search and user relevance feedback into a unified search flow, replacing the previous approach where each was isolated in separate panes, requiring the user to mediate between the two subsystems. The new approach utilizes rank aggregation to combine results from both methods and offers varying display options in the interface. Additionally, we have introduced the capability to use multiple models and redesigned the video summary browser, allowing users to view entire videos more easily rather than navigating segment by segment. We believe these changes will enhance users’ retrieval capabilities for various tasks involving large multimedia collections.

Acknowledgments. This work was supported by Icelandic Research Fund grant 239772-051.

References

1. Cherti, M., et al.: Reproducible scaling laws for contrastive language-image learning. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pp. 2818–2829 (2023)
2. Ilharco, G., et al.: OpenCLIP (2021)
3. Ro, Y.M., et al. (eds.): MMM 2020. LNCS, vol. 11961. Springer, Cham (2020). <https://doi.org/10.1007/978-3-030-37731-1>
4. Khan, O.S., et al.: Exquisitor at the video browser showdown 2021: relationships between semantic classifiers. In: Lokoč, J., et al. (eds.) MMM 2021. LNCS, vol. 12573, pp. 410–416. Springer, Cham (2021). https://doi.org/10.1007/978-3-030-67835-7_37
5. Khan, O.S., et al.: Interactive learning for multimedia at large. In: Jose, J.M., et al. (eds.) ECIR 2020. LNCS, vol. 12035, pp. 495–510. Springer, Cham (2020). https://doi.org/10.1007/978-3-030-45439-5_33
6. Khan, O.S., et al.: Exquisitor at the Video Browser Showdown 2022. In: Proceedings of Multimedia Modeling (MMM), pp. 511–517. Springer (2022)
7. Khan, O.S., Sharma, U., Zhu, H., Rudinac, S., Jónsson, B.T.: Exquisitor at the lifelog search challenge 2024: Blending conversational search with user relevance feedback. In: Proceedings of the 7th Annual ACM Workshop on the Lifelog Search Challenge, p. 117121. LSC '24, Association for Computing Machinery (2024)
8. Khan, O.S., Zhu, H., Sharma, U., Kanoulas, E., Rudinac, S., Jónsson, B.Þ: Exquisitor at the video browser showdown 2024: relevance feedback meets conversational search. In: Rudinac, S., et al. (eds.) MultiMedia Modeling, pp. 347–355. Springer Nature Switzerland, Cham (2024)
9. Kratochvíl, M., Veselý, P., Mejzlík, F., Lokoc, J.: SOM-hunter: video browsing with relevance-to-SOM feedback Loop. In: Proceedings of MultiMedia Modeling (MMM). Daejeon, South Korea (2020)
10. Lokoč, J., et al.: Interactive video retrieval in the age of effective joint embedding deep models: lessons from the 11th VBS. Multimedia Systems, pp. 1–24 (2023)
11. Lokoč, J., et al.: Interactive search or sequential browsing? A detailed analysis of the video browser showdown 2018. ACM Trans. Multimedia Comput. Commun. Appl. **15**(1) (2019)
12. Radford, A., et al.: Learning transferable visual models from natural language supervision. In: International Conference on Machine Learning, pp. 8748–8763. PMLR (2021)
13. Rossetto, L., Gasser, R., Sauter, L., Bernstein, A., Schuldt, H.: A system for interactive multimedia retrieval evaluations. In: Lokoč, J., et al. (eds.) MultiMedia Modeling, pp. 385–390. Springer International Publishing, Cham (2021)
14. Rossetto, L., Schuldt, H., Awad, G., Butt, A.A.: V3C - a research video collection. In: Proceedings of MultiMedia Modeling (MMM). Thessaloniki, Greece (2019)
15. Rui, Y., Huang, T.S., Mehrotra, S.: Content-based image retrieval with relevance feedback in mars. In: Proceedings of International Conference on Image Processing (ICIP) (1997)

16. Schoeffmann, K.: Video browser showdown 2012-2019: a review. In: 2019 International Conference on Content-Based Multimedia Indexing (CBMI), pp. 1–4 (2019)
17. Truong, Q.T., et al.: Marine video kit: a new marine video dataset for content-based analysis and retrieval. In: Dang-Nguyen, D.T., et al. (eds.) MultiMedia Modeling, pp. 539–550. Springer International Publishing, Cham (2023)